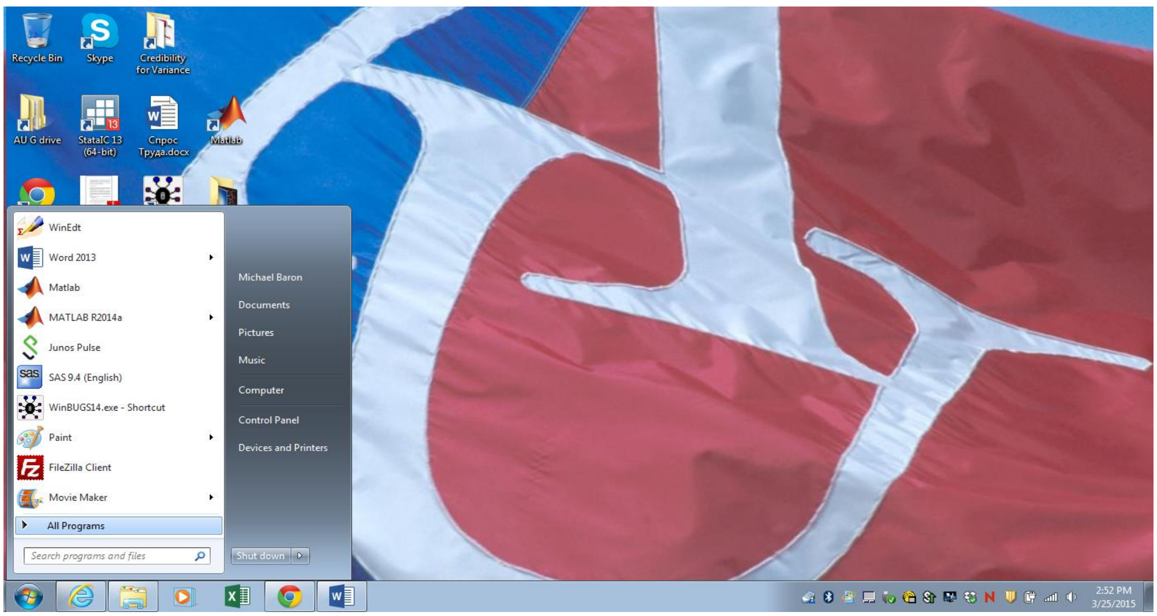
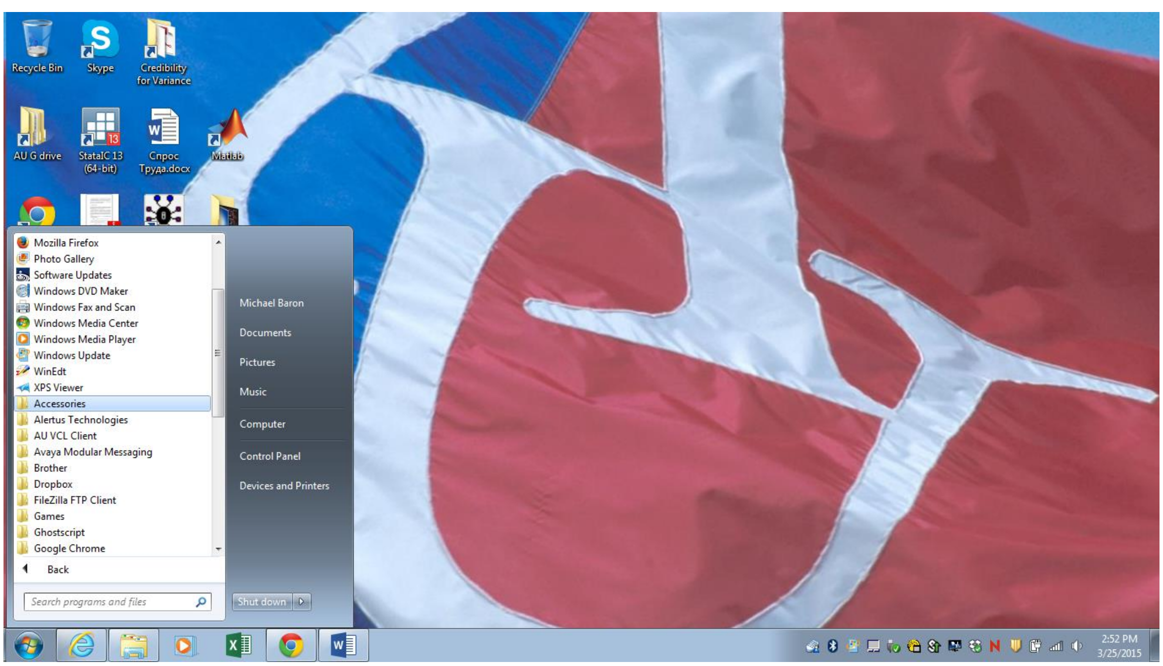
<u>BUGS Software (Bayesian Inference Using Gibbs Sampling)</u>

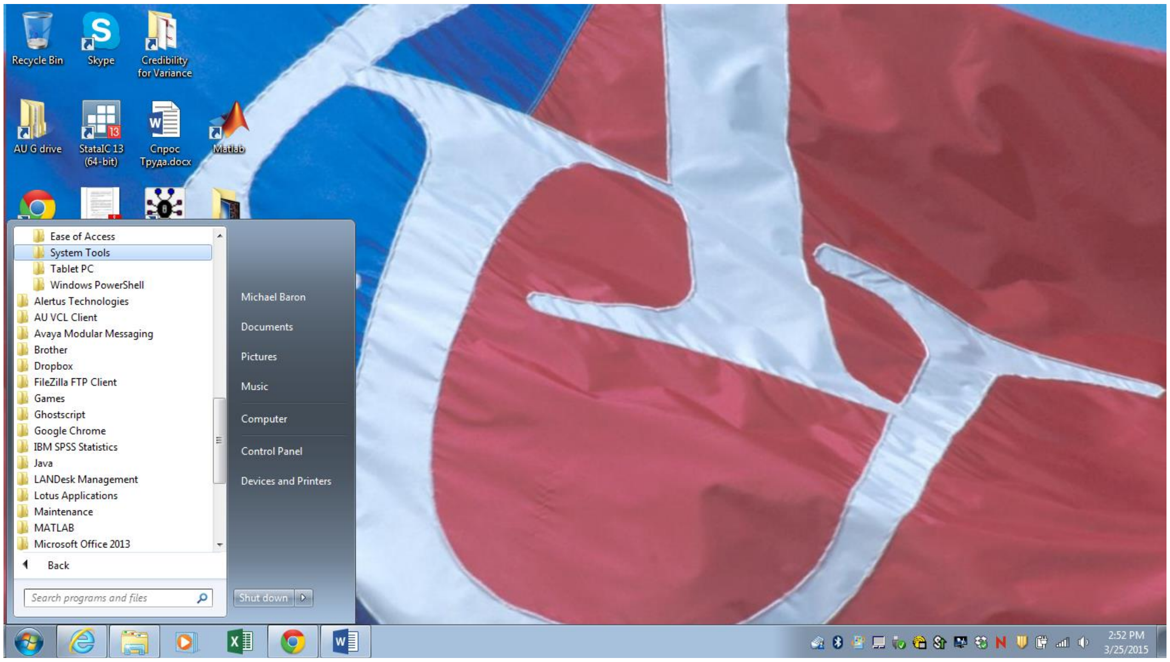
1. <u>Installation</u>

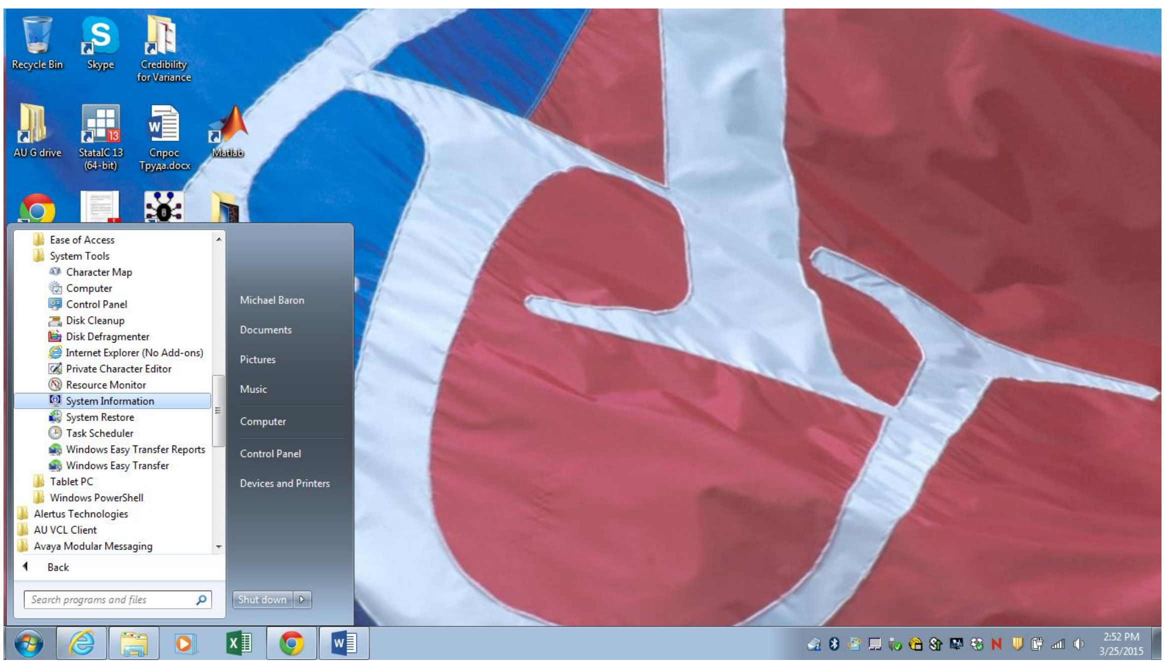
<u>Step 1. Preparation.</u> Find out whether you are running a 32-bit or a 64-bit machine. The answer is in “System information”:

Go to Start → Programs → Accessories → System Tools → System Information






Recycle Bin
Skype
Credibility for Variance
AU G drive
StataIC 13 (64-bit)
Спрос Труда.docx
Matlab
Ease of Access
System Tools
Tablet PC
Windows PowerShell
Alertus Technologies
AU VCL Client
Avaya Modular Messaging
Brother
Dropbox
FileZilla FTP Client
Games
Ghostscript
Google Chrome
IBM SPSS Statistics
Java
LANDesk Management
Lotus Applications
Maintenance
MATLAB
Microsoft Office 2013
Back
Search programs and files
Michael Baron
Documents
Pictures
Music
Computer
Control Panel
Devices and Printers
Shut down
2:52 PM
3/25/2015


Recycle Bin
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System Tools
Character Map
Computer
Control Panel
Disk Cleanup
Disk Defragmenter
Internet Explorer (No Add-ons)
Private Character Editor
Resource Monitor
System Information
System Restore
Task Scheduler
Windows Easy Transfer Reports
Windows Easy Transfer
Tablet PC
Windows PowerShell
Alertus Technologies
AU VCL Client
Avaya Modular Messaging
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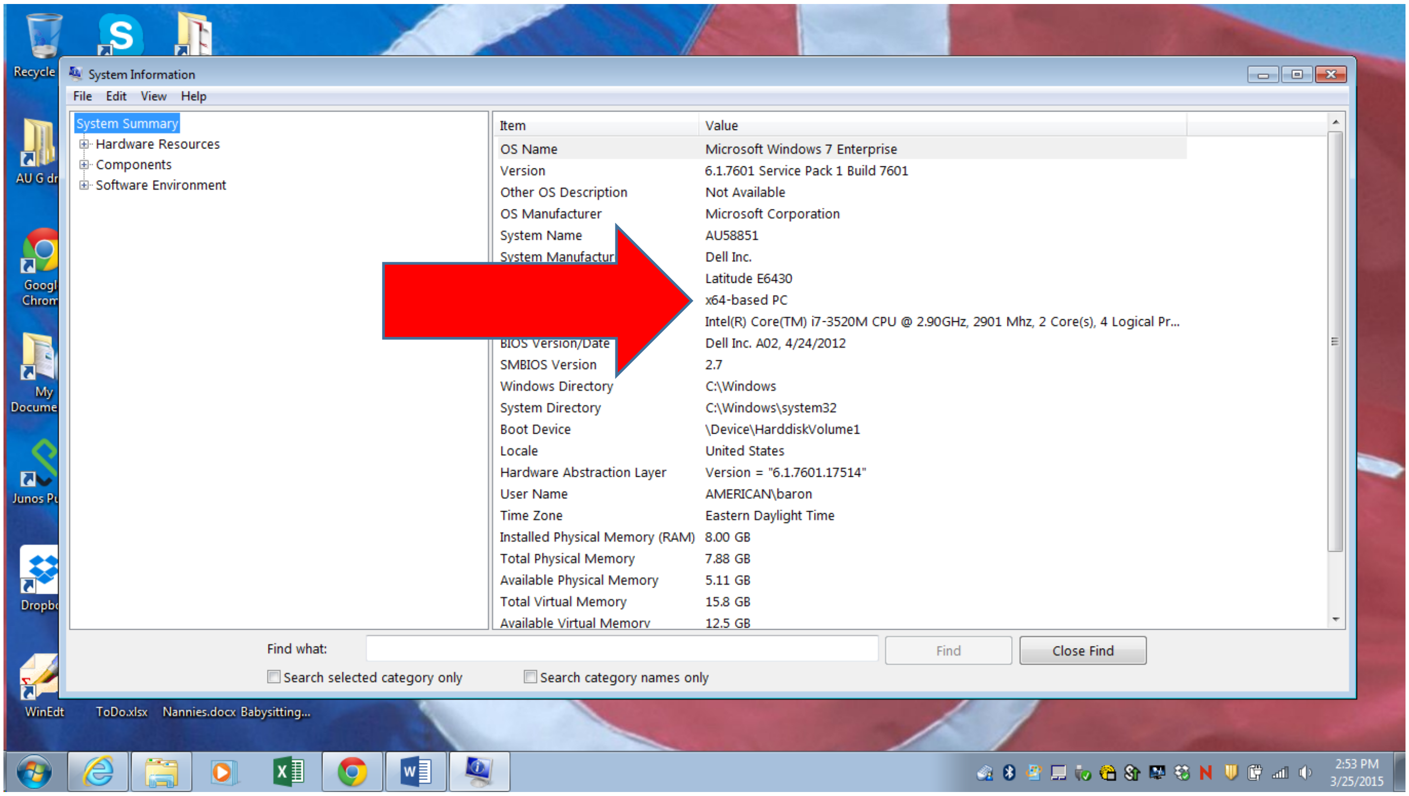
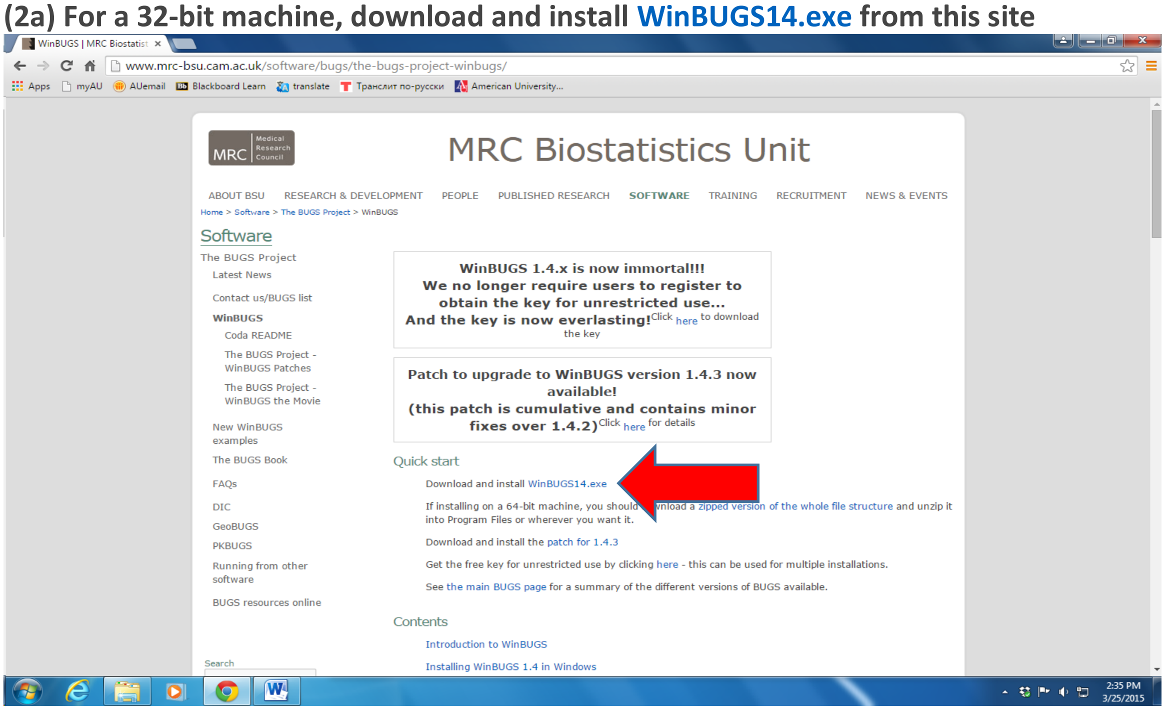
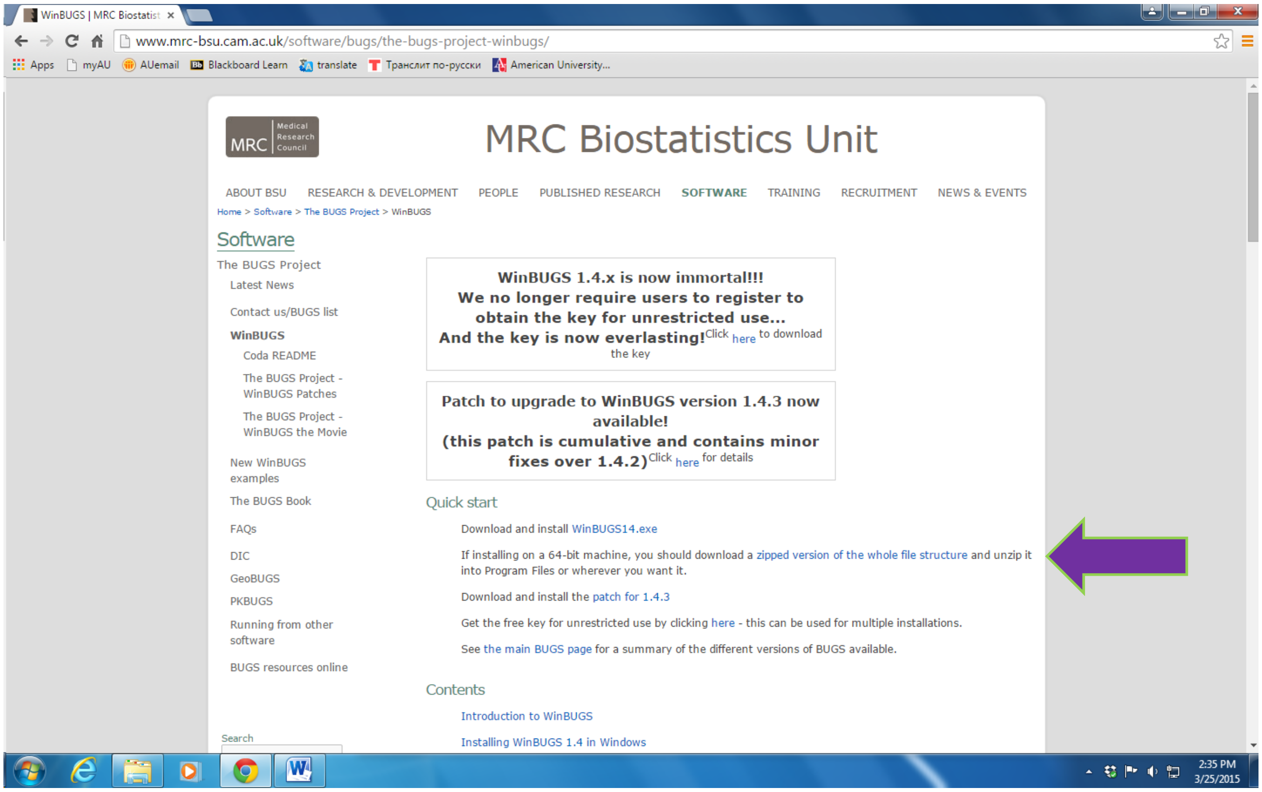


<u>Step 2.</u> WinBUGS, now a free version of BUGS, is available from http://www.mrc-bsu.cam.ac.uk/software/bugs/the-bugs-project-winbugs/

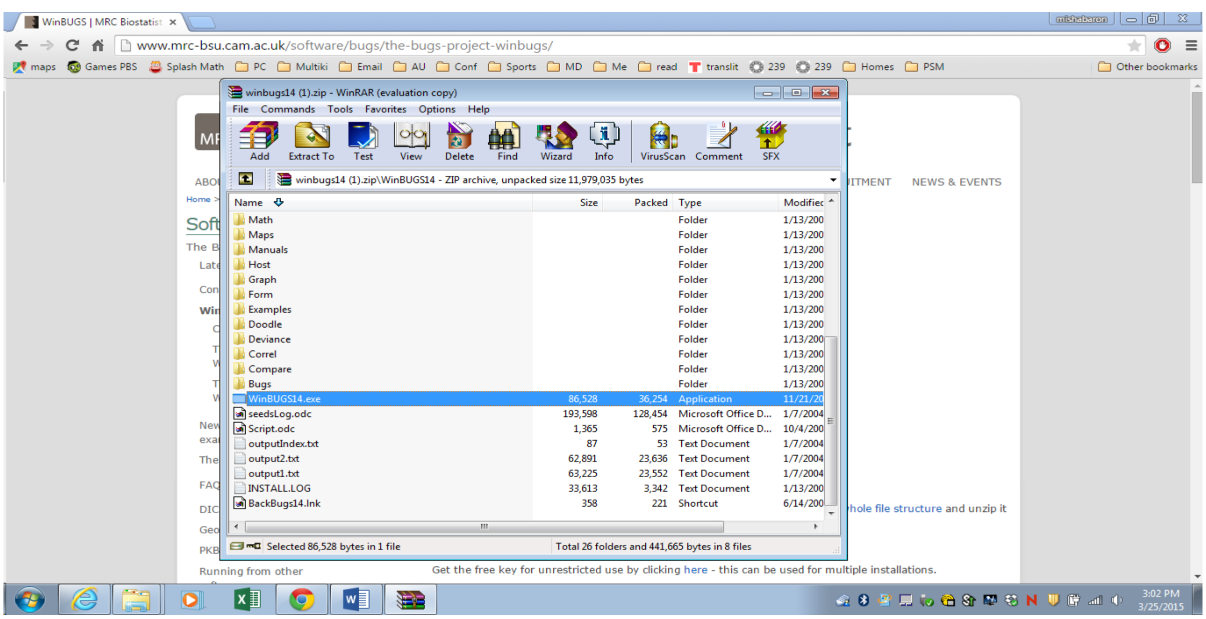
(2a) For a 32-bit machine, download and install WinBUGS14.exe from this site



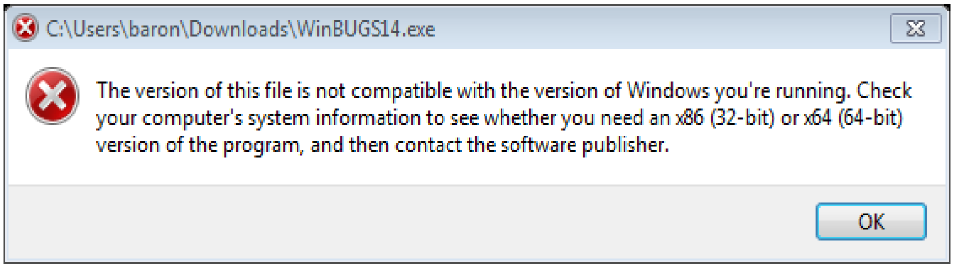
(2b) For a 64-bit machine, download a zipped version …



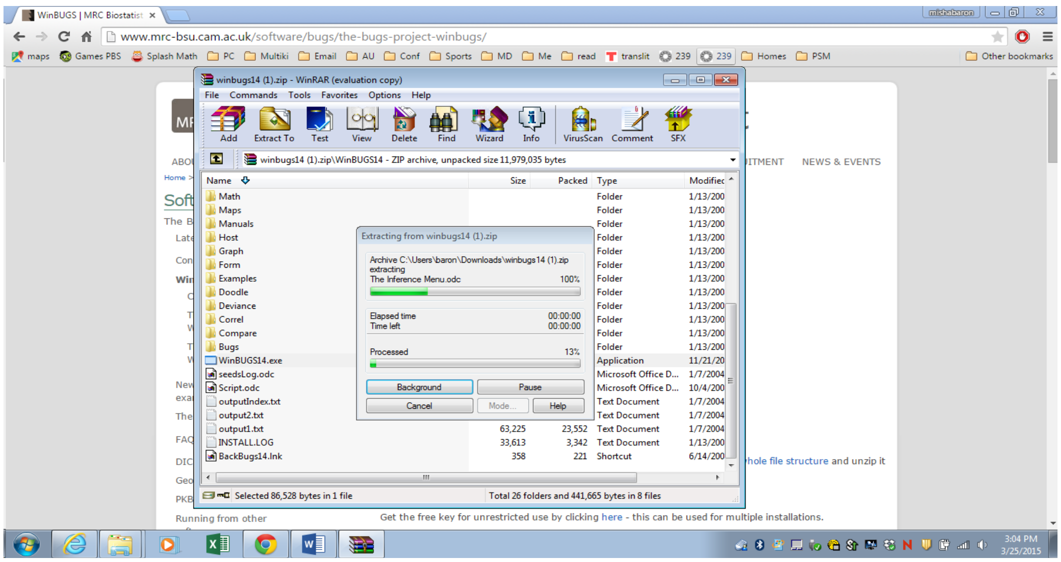
… and unzip it.



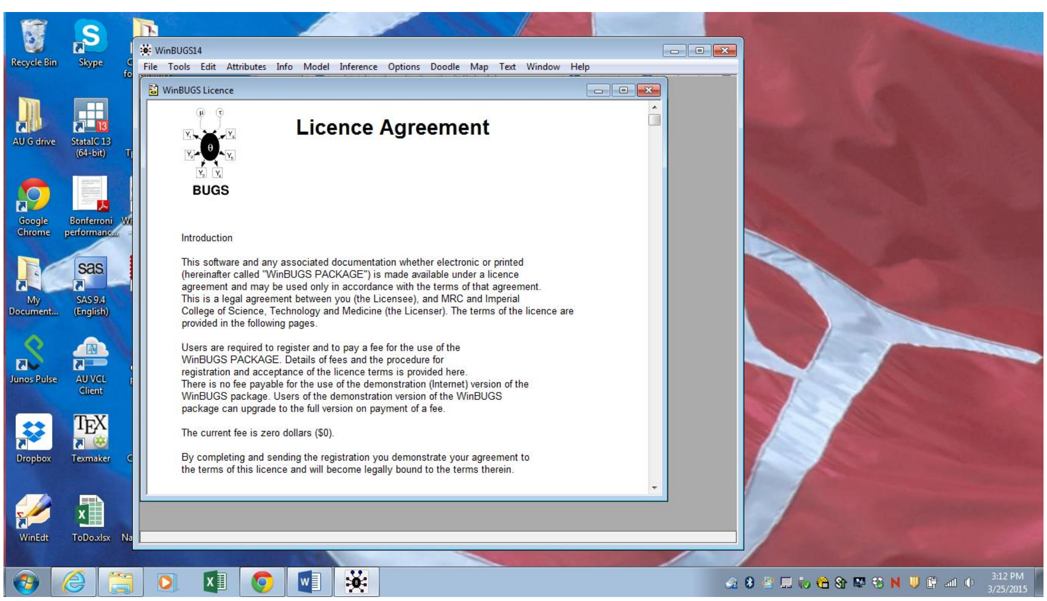
If you are not sure whether you have a 32-bit or 64-bit computer, and you try a wrong version, the system will complain:



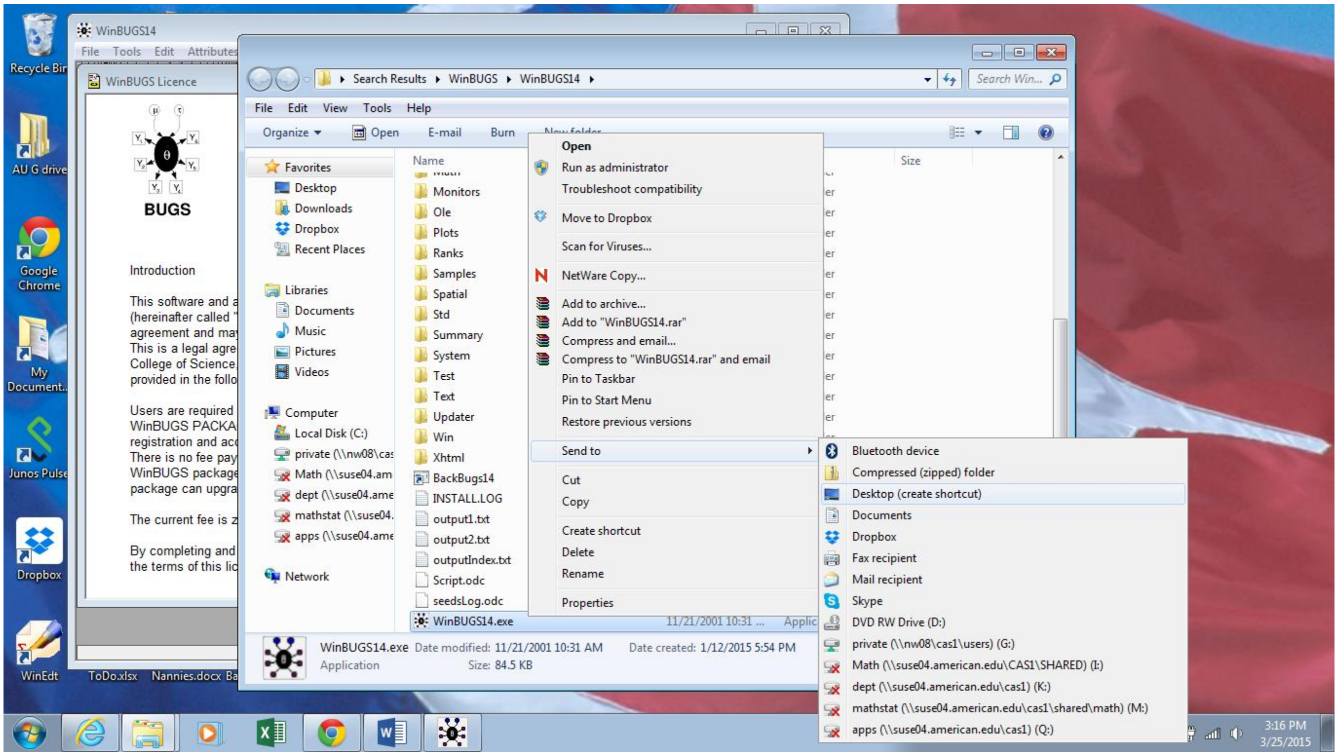
Execute the file WinBUGS14.exe



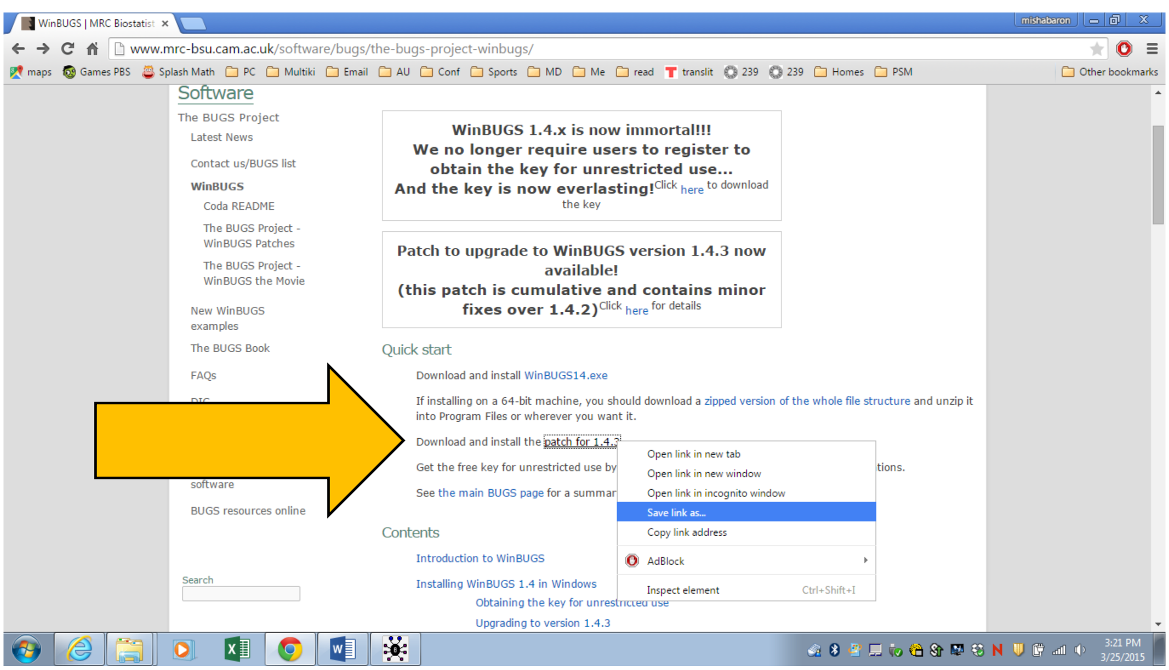
<u>WinBUGS opens:</u>



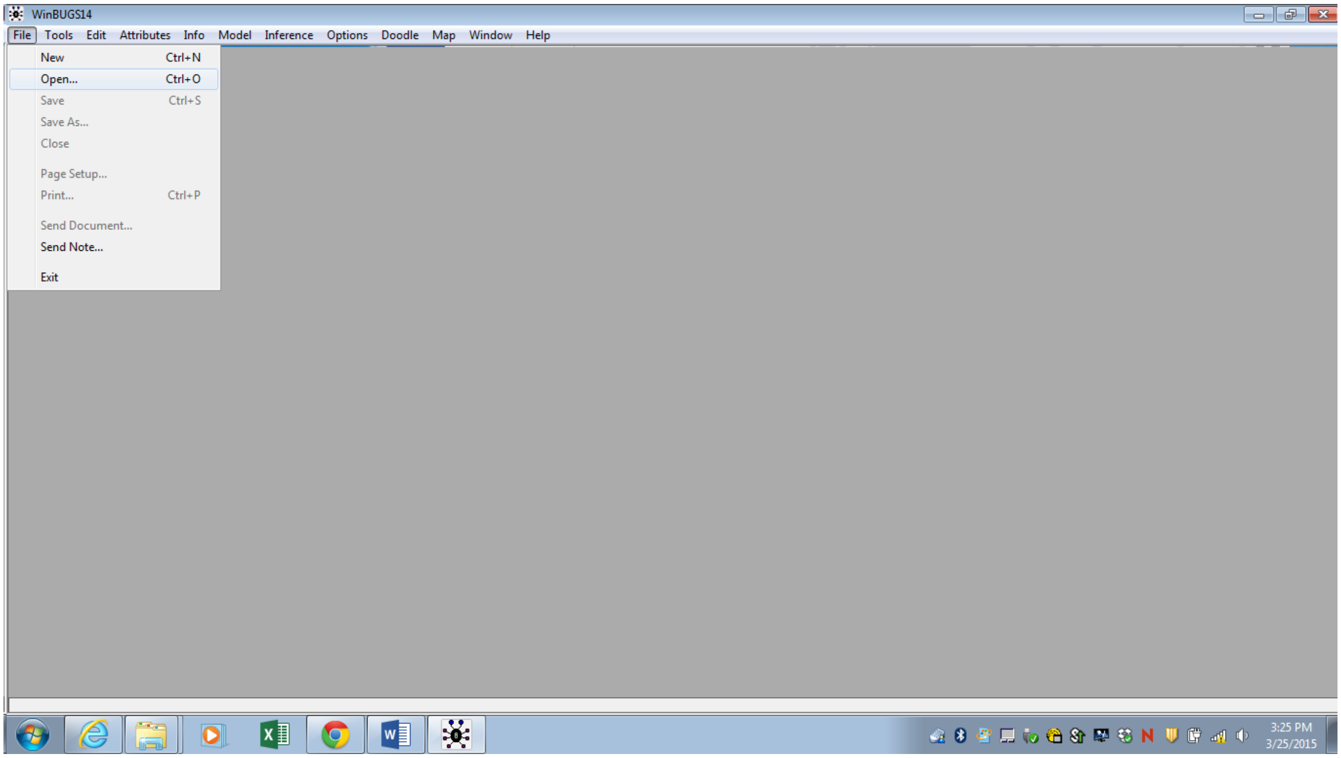
Create a shortcut of WinBUGS.exe to desktop for your convenience by sending the file to desktop (create shortcut).

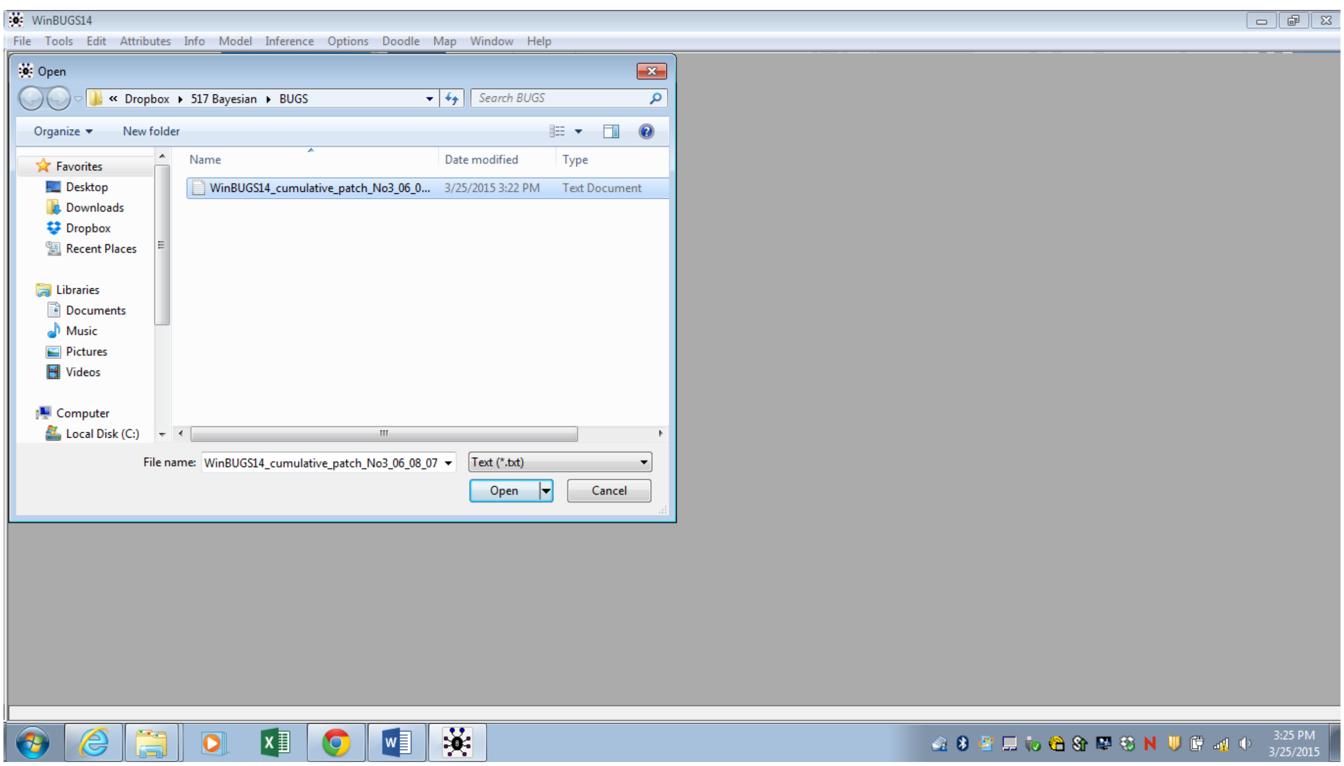


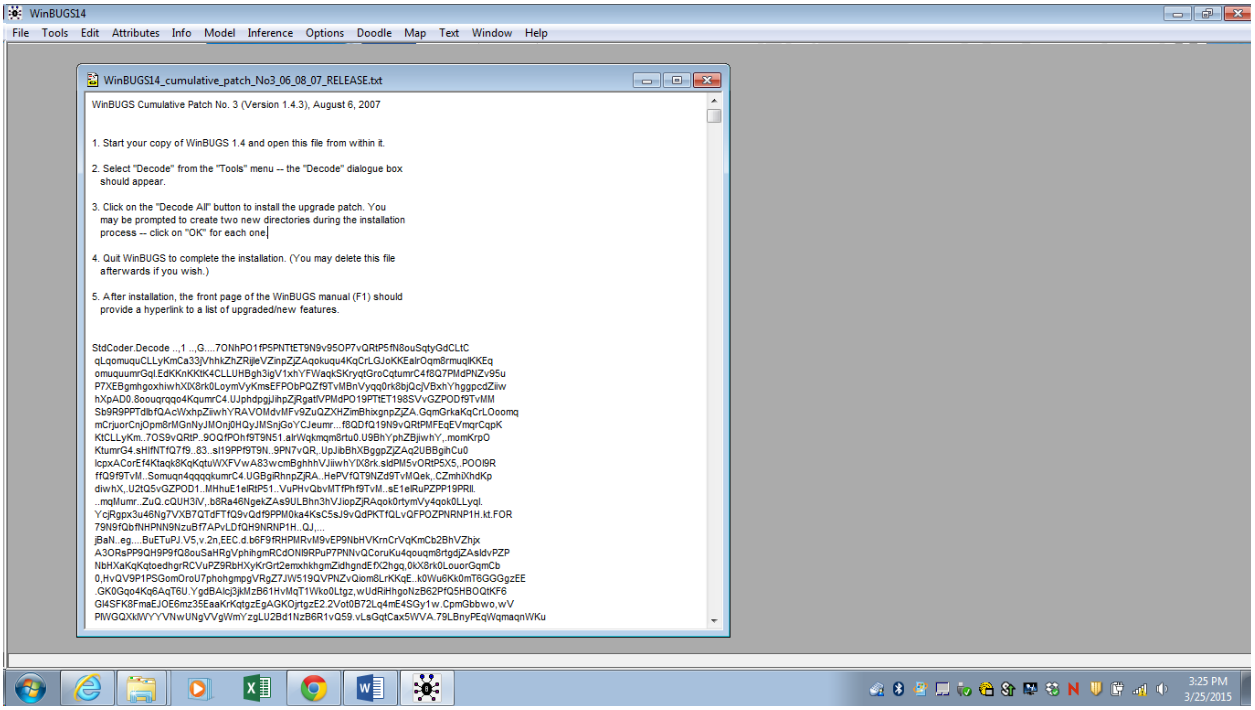
<u>Step 3. Registration for eternal free use.</u> On the same site, find "Download and install the patch for 1.4.3", right-click on it, save the txt file, then open it in WinBUGS.



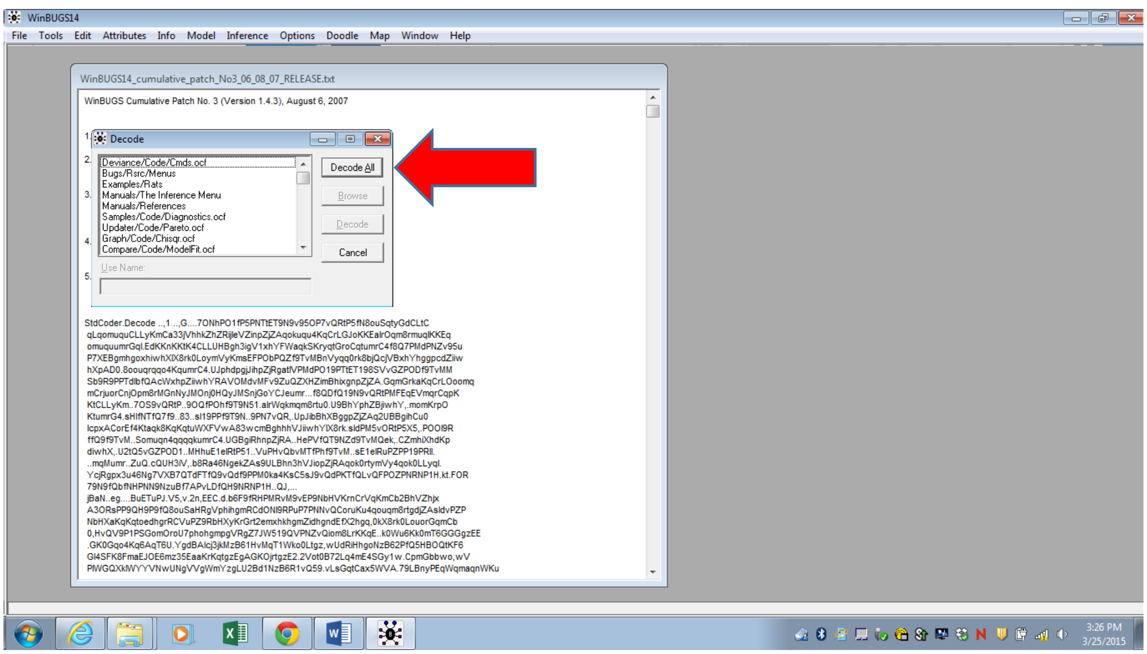
In WinBUGS, go to File → Open, and open this text file with the patch.



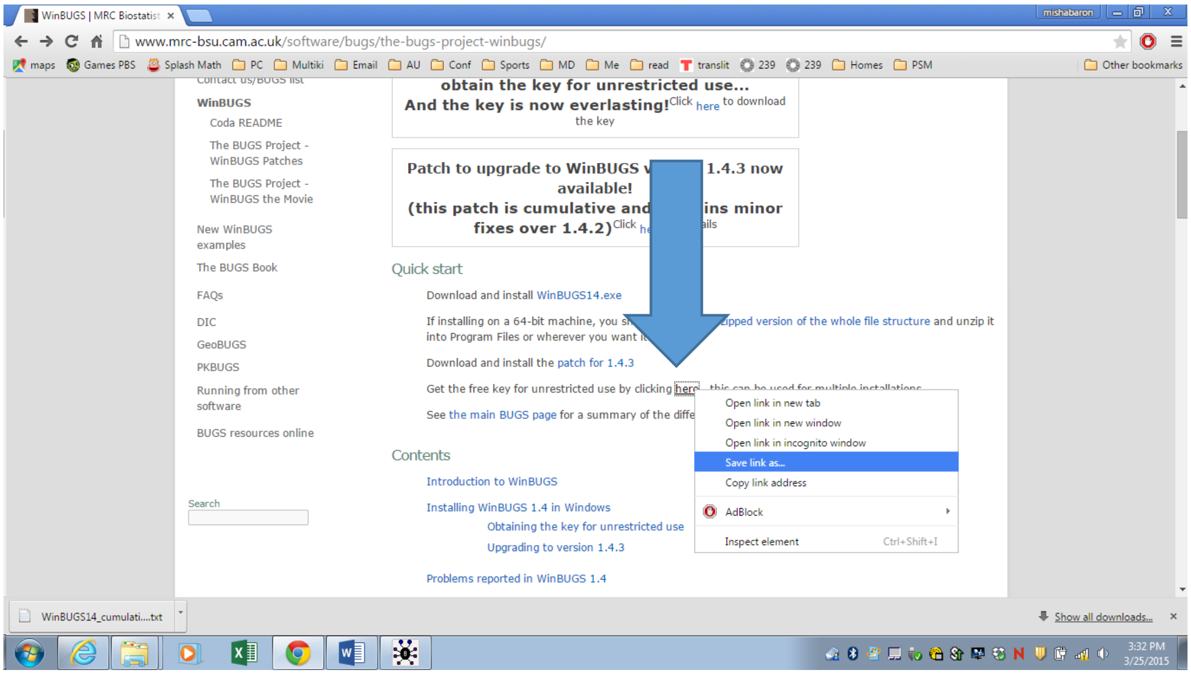


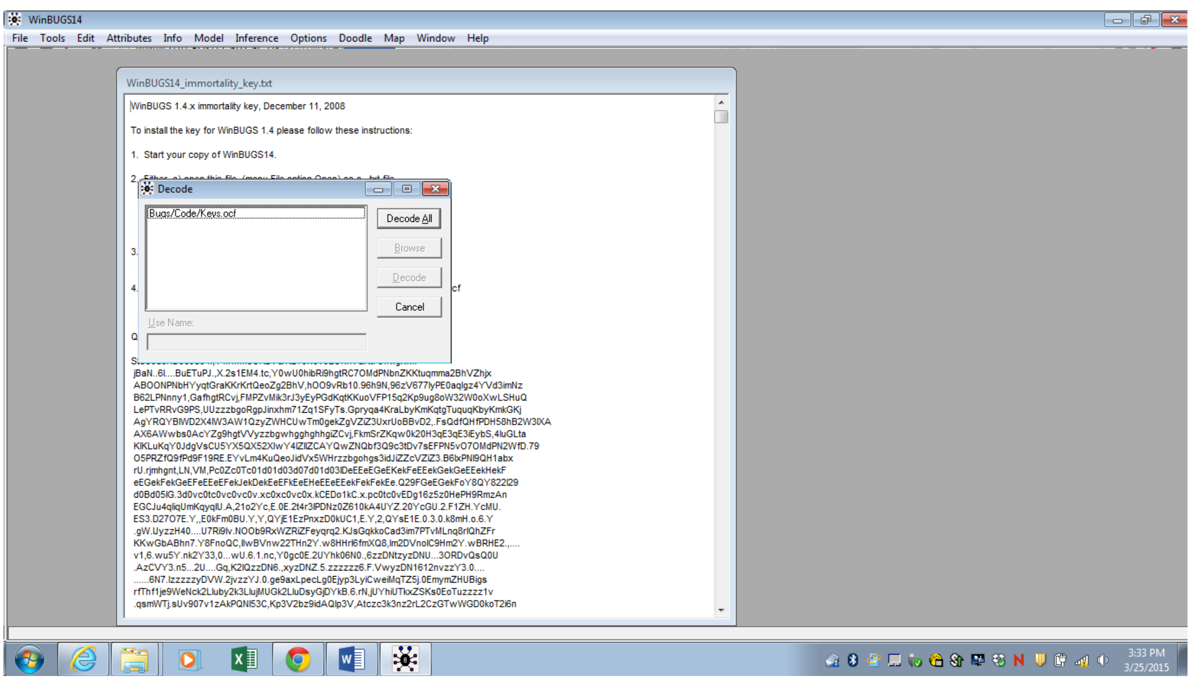


Go to Tools → Decode, then press "Decode all".



From the same site, get the "immortality key" for unrestricted use and do the same with it - save as txt file, open it in WinBUGS, Decode, and choose "Decode all".





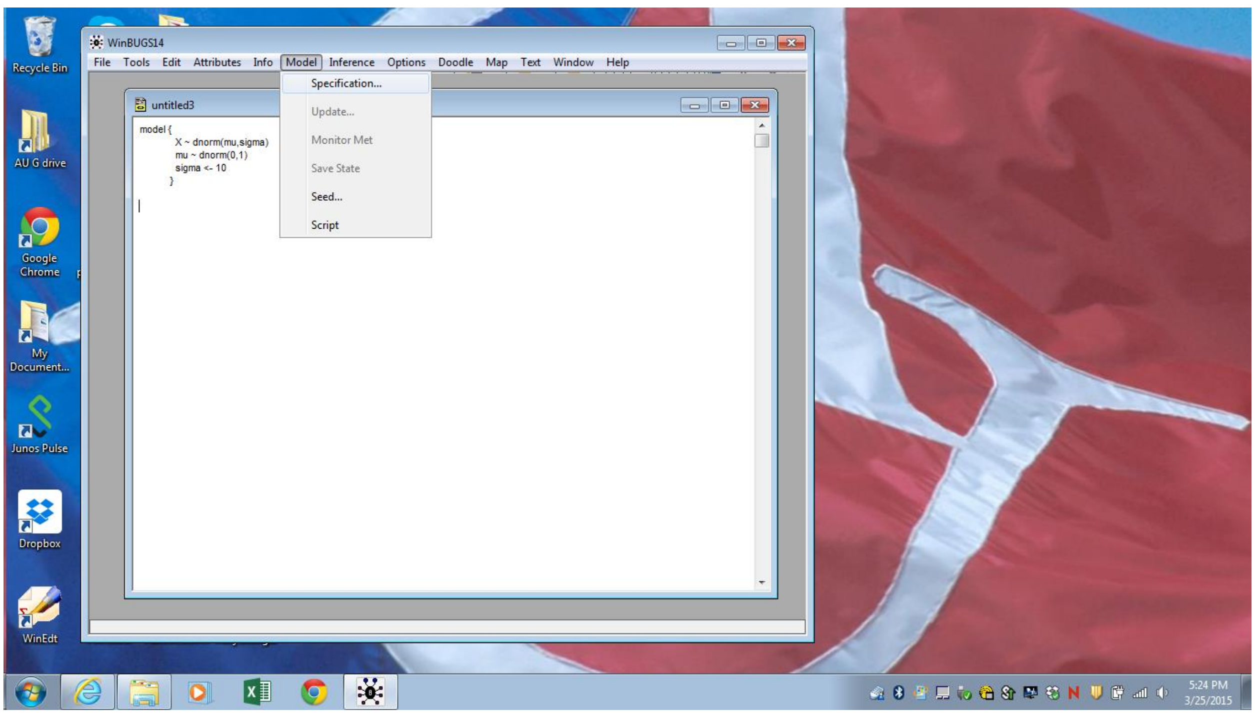
2. Basic Steps

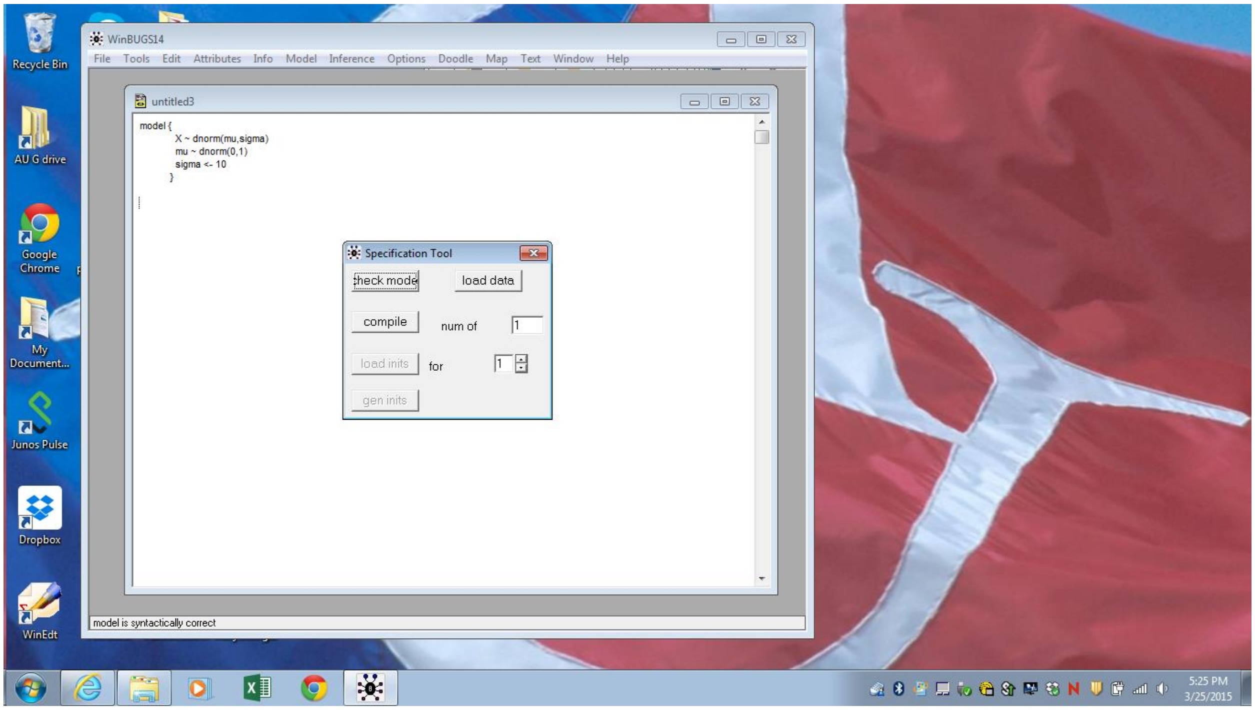
Write a program

```
model  {
        X ~ dnorm(mu,sigma)
        mu ~ dnorm(0,1)
        sigma <- 10
}
```

Verify it and compile:

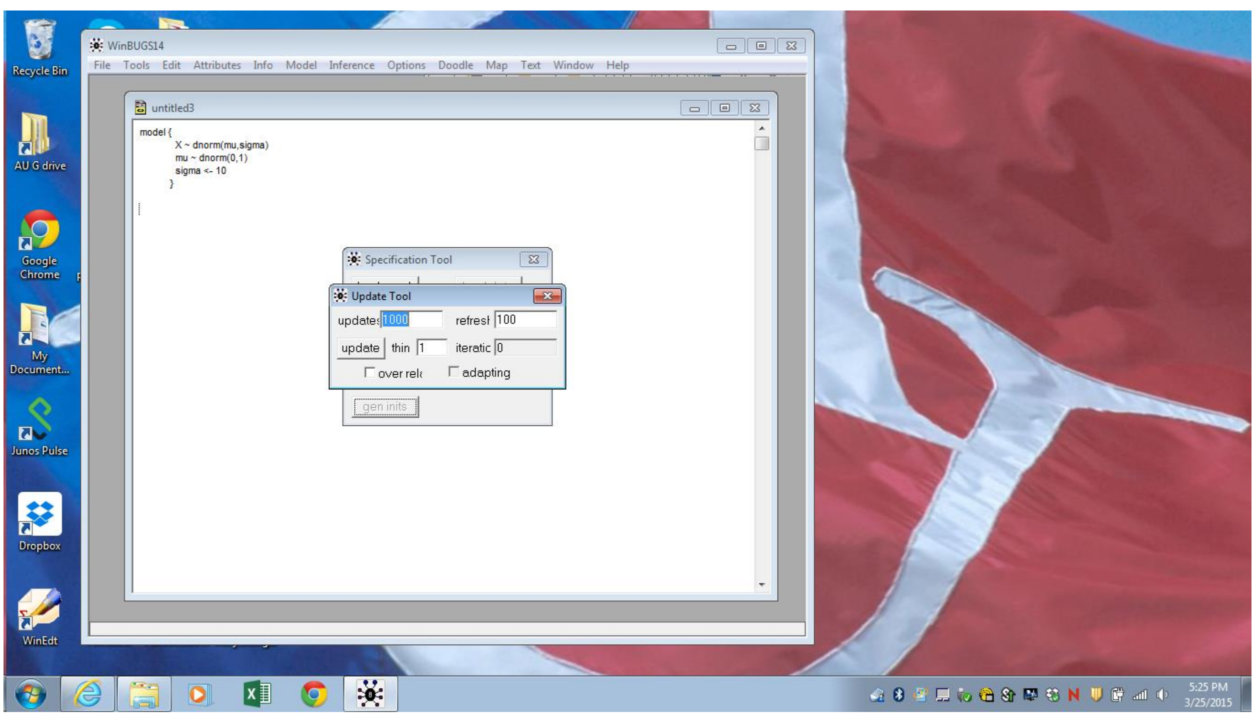
Model -> specification tool -> check model   Response: synthetically correct





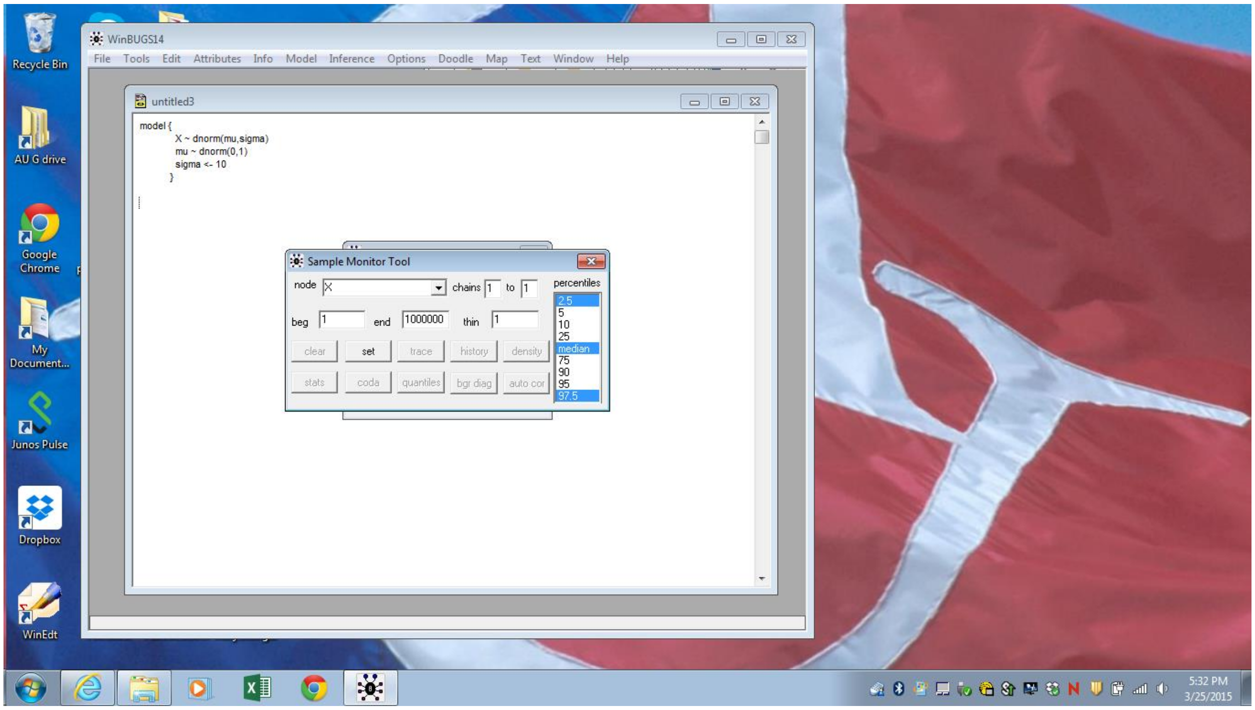
- ➔ Compile Response: compiled
- ➔ Gen inits Response: model initialized

Model -> update



Specify variables we want to monitor.

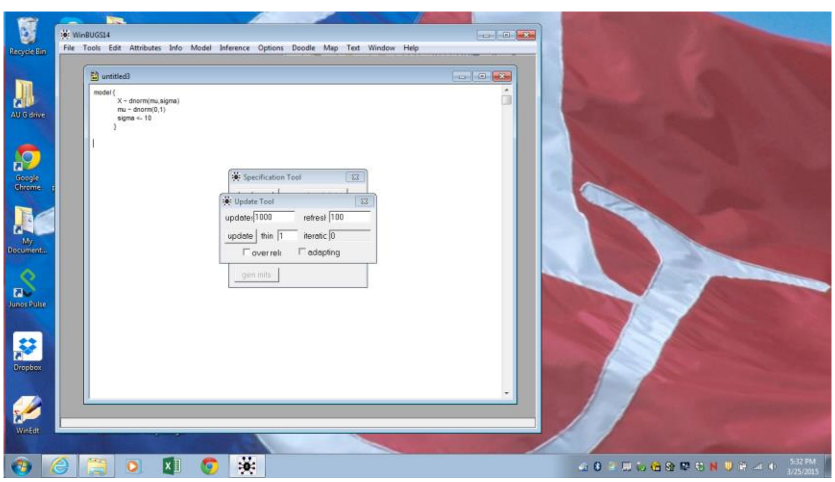
Inference -> Samples



- ➔ node -> X -> set
- ➔ node -> mu -> set
- ➔ ……………………..
- ➔ node -> * -> trace

Run the simulations.

Update tool window -> update



Collect the desired statistics and observe the desired graphs for the monitored variables

Sample Monitor window -> stats
Sample Monitor window -> trace
Etc.